1.List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Difference Between IPv4 and IPv6

IPv4	IPv6
1. Uses 32-bit addresses.	Uses 128-bit addresses.
2. Example: `192.168.1.1`	Example: `2001:db8::1`
3. Supports about 4.3 billion addresses.	Supports a very large number of addresses.
4. Written in numbers separated by dots.	Written in hexadecimal numbers separated by colons.
5. Example use: Older routers and networks.	Example use: Modern networks and devices.
6. NAT is commonly used to share one public IP address among many devices.	NAT is generally not required because IPv6 provides many addresses.

2.Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Static IP Address Table (Easy Answer)

We have 4 devices connected to home Wi-Fi. We can assign a different static IP address to each device using the `192.168.10.0/24` network.

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

3.Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Answer: Subnetting with /27 CIDR

The network is `192.168.1.0` and the subnet mask is `/27`.

- Subnet mask: 255.255.255.224
- Number of subnets: 8 subnets (when dividing a /24 network into /27 subnets)
- Hosts per subnet: 30 usable hosts
- Block size: 32 IP addresses per subnet

First 3 Subnet Ranges

Subnet	Network Address	Usable Host IPs	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

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**4.You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.**

Answer (Easy Way)

Given:

- IP Address: `10.0.5.17`
- Subnet Mask: `255.255.255.0`
- CIDR Notation: `/24`

Item	Answer
Network Address	`10.0.5.0`
Broadcast Address	`10.0.5.255`
First Valid Host	`10.0.5.1`
Last Valid Host	`10.0.5.254`
Total Usable Hosts	254

5.Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Verify the Subnet Calculation

Steps to follow:

1. Open the Subnet Calculator.
2. Enter IP address: `10.0.5.17`
3. Enter subnet mask: `255.255.255.0` (or `/24`).
4. Click Calculate if needed.
5. Take a screenshot of the results and attach it to your assignment.

Expected Results

Field	Result
Network Address	`10.0.5.0`
Broadcast Address	`10.0.5.255`
First Valid Host	`10.0.5.1`
Last Valid Host	`10.0.5.254`
Usable Hosts	`254`